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Digital Object Identifier

Physics-Based Event Attribution From Sparse PMUs With Recovery Memory

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT An electrical network can continue oscillating after a load is restored or a line is reclosed. A later intervention can then resemble the recovery of an earlier one. We address this ambiguity by comparing physical histories from eight sparse PMUs. DAE-derived switched predictors use voltage, oriented terminal current, frequency and ROCOF, start from estimated past states, and preserve recovery through restoration. A ten-second state history supports screened onset proposals; conditional evidence integrates load severity and updates fixed network hypotheses. Active-event and originating-cause masses keep ongoing interventions separate from their remaining response. Completed development replays with a common publication clock give 51/54 timely family/source decisions with 3 extra alarms, versus 37/54 and 37 for a simpler state-retaining physical procedure. Independent confirmation is in progress. The contribution is a specified recovery-aware attribution mechanism and an evaluation separating detection, timely location and restoration. The frozen physical startup contract rejects the retrospective RAW reference; a separate learned source hybrid corrects only one additional physical window. The supported scope is a known load/line catalog; probability calibration, all-family diagnosis and real-time deployment remain unestablished.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.[1]

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [2]; physical outage detectors compare network hypotheses sequentially [3], [4]; and simulated fault trajectories support diagnosis from sparse PMUs [5]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.

[1]Drafting/revision assistance: OpenAI Codex [1].

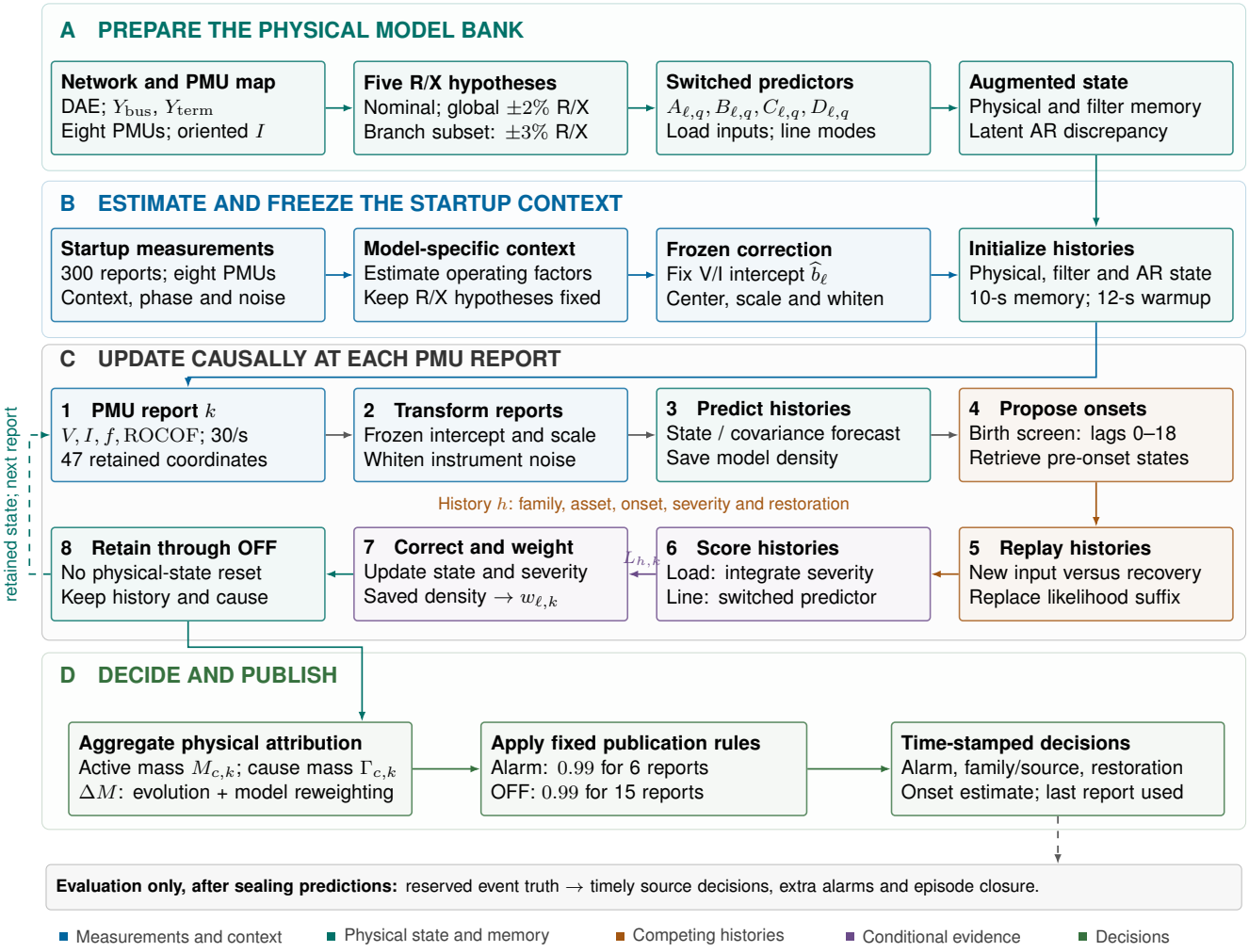


FIGURE 1. Implemented physical-history attribution with its original publication policy, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed. The common-clock controls replace block D as specified in Section VII.

2) A comparison that separates detection, correct source attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.
3) Adversarial tests of scope: a state-retaining physical competitor, a classifier fitted under a grouped SIM split, and a retrospective integrity/source replay. These tests retain failures and distinguish simulation utility from transfer to the reference record.

A further physical control asks whether recovery increases source separation. It distinguishes retaining a consistent prediction from obtaining additional information about the source; it is not a comparison of causal detectors.

On the common development cohort and publication clock, the five-model bank obtains 51/54 timely family/source decisions with 3 extras, versus 37/54 and 37 for the simpler physical control. This compares complete procedures; it does not isolate a causal effect of history retention. The prespecified confirmatory comparison is still running without method changes. The claim is scoped to the declared load/line catalog. These simulations do not establish an all-family hackathon score, calibrated event probabilities or real-time deployment.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [2], [6], [7]. Functional observability asks whether a requested state quantity can be recovered [8], [9]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [10], [11].

Physical hypotheses support outage inference from phasor

angles [12], Bayesian inference with uncertain states [13], and sequential detection under stochastic operation [3], [4]. Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics [14]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [15]–[17].[2]

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [5]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [18]. Multiple-model filtering already propagates conditional state/mode hypotheses [19]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [20]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [21]–[23]. We compare two fixed ExtraTrees adapters from the SGSMA project. Their performance tests transfer of those implementations, not the limits of all supervised learning. A known physical catalog and source-excluded supervised training also impose different information constraints.

Physical stochastic models support latent-state estimation [24], [25], while joint DAE parameter inference addresses uncertain generators and networks [26]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [27]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

[2]Drafting/revision assistance: OpenAI Codex [1].

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.[3]

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

[3]Drafting/revision assistance: OpenAI Codex [1].

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

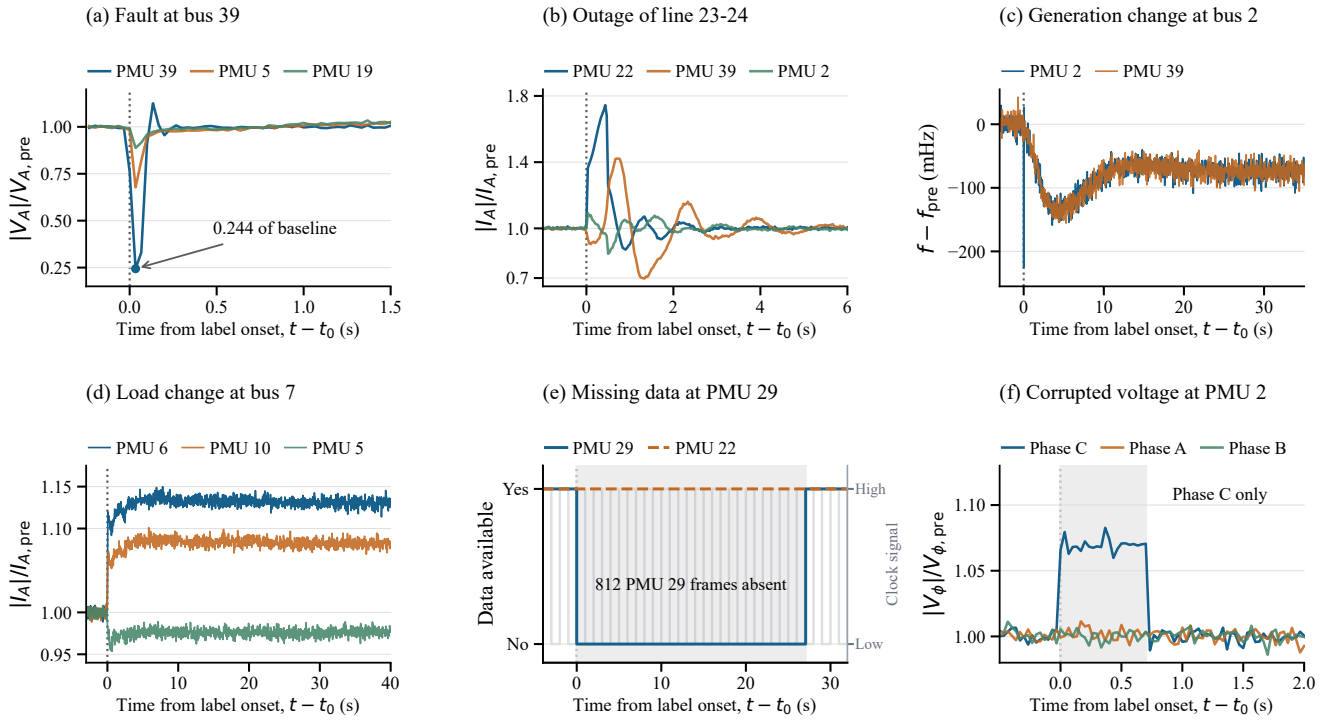


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine, archived as R11, uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.[4]

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insuffi-

[4]Drafting/revision assistance: OpenAI Codex [1].

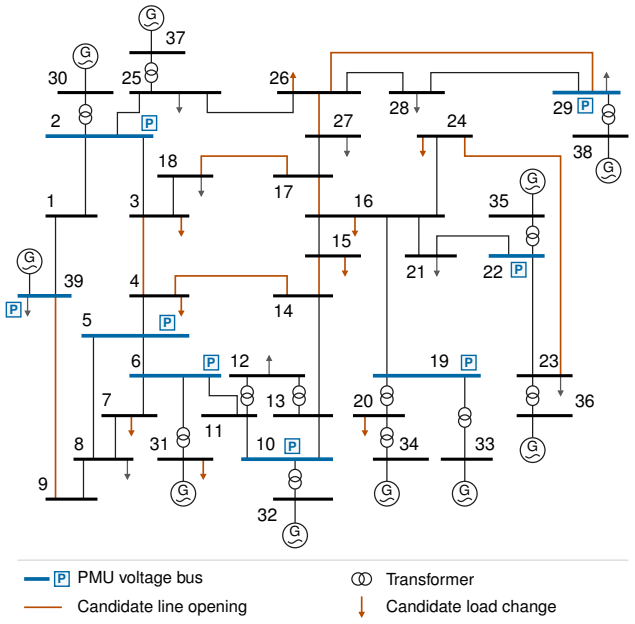


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

cient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u), \\ \delta\dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \overline{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

The supplement derives the closed form by completing the square and works through a numerical example. It also specifies the conditional state representation $m_h + d_h a$ and its update. The local amplitude integral is exact for the stated conditional Gaussian model. The complete bank remains approximate: regime covariances are shared from nominal-operator Riccati equations, context estimates enter by substitution, and histories are screened and pruned. At episode closure the augmented state mixture mean is retained and omitted between-history spread is recorded. A fully Bayesian posterior over states, context and all histories is not claimed.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{11}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{12}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{13}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\begin{aligned} \Delta M_{c,k} = &\sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ &+ \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \tag{14}$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The supplement records the frozen birth/restoration hazards, severity prior and integration intervals, restoration-child rules and beam limits, together with the execution sequence and illustrative calculations. These details preserve the archived configuration; the compact algorithm below states the report order and information boundary.

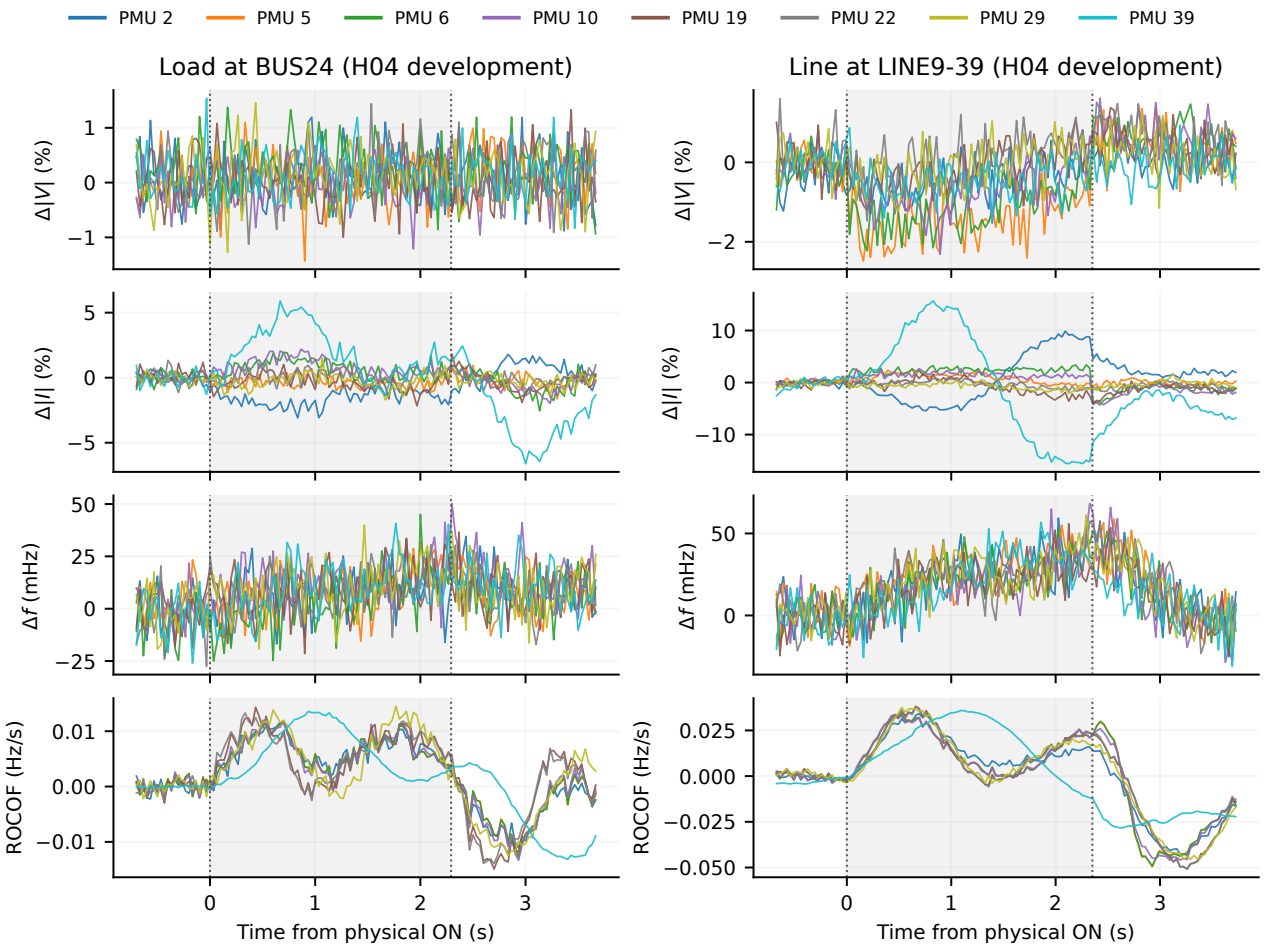


FIGURE 4. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.[5]

The simulator is the IEEE-39 Julia/PowerDynamics research candidate F01. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. The two archived ExtraTrees artifacts (the R09 references) and their preprocessing/publication adapters are also frozen; neither is retrained on H04 or C01. They receive the same 30-fps observations. The separately fitted S01 SHORT classifier uses the grouped SIM contract described below and is then frozen for H04 and RAW. It does not replace the archived comparators in C01. These are comparisons of specified implementations, not a hackathon leaderboard rerun.

The published physical comparator is Chen *et al.*'s persistent-line quickest-change detector [3], which considers single- and double-line outages. Our declared F01 adapter restricts its fixed nominal catalog to single-line hypotheses. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. The adapter uses nonoverlapping angle increments at 15 Hz and a startup-only scale/drift estimate. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign, C01, therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent evaluation is in progress. The following section reports completed development results only.

VI. CONTINUOUS-STREAM RESULTS

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

This section uses each frozen method's original publication policy. The common-clock physical control in Section VII republishes the same history-bank outputs under a different declared rule; its counts are a separate endpoint. The ExtraTrees rows here are the archived R09 references, distinct from the newly fitted S01 classifier.[6]

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 4). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multiclass hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 6 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later

[5]Drafting/revision assistance: OpenAI Codex [1].

[6]Drafting/revision assistance: OpenAI Codex [1].

TABLE 3. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Signature component test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Recovery geometry control	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; not R11 performance
H04 procedure controls	Same development parents and observations as above	Single-state GLR, history banks and newly fitted ET	Common event metrics; distinct score/publication semantics
Grouped SIM learning	31/8/17 fit/validation/reserved groups; 90/16/70 records	Frozen classifier selection and reserved performance	Previously open campaigns; no unseen-source claim
RAW reference replay	One retrospective record; 180 fixed windows, 24 physical-positive	Physical admission, integrity timing and source composition	Metadata-based identities; no exact actuation log or independent field claim

TABLE 4. H04 with the original publication policies. ExtraTrees rows are archived R09 comparators. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
Archived ET ref.	0/27	27/27	3	0
Archived ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (14) separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative experiment leaves R11 and nominal frozen for C01.

VII. ADVERSARIAL CONTROLS AND THE TRANSFER BOUNDARY

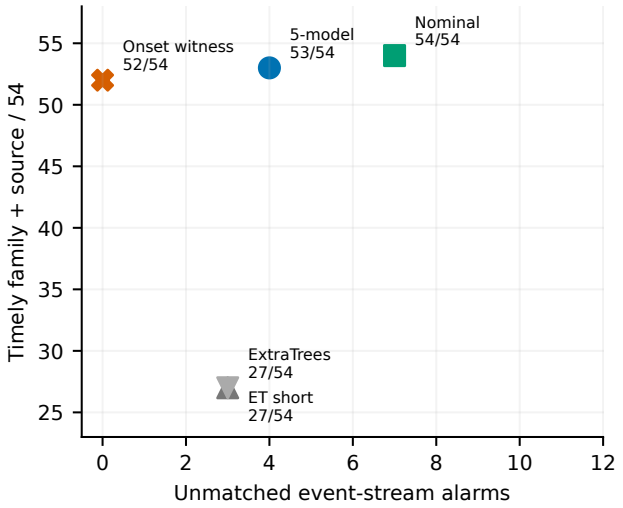


FIGURE 5. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

A. A SIMPLER PHYSICAL COMPETITOR

A stronger simplicity check retains one committed physical and instrumental state, rather than resetting to equilibrium after each event. Short GLR alternatives test the same admitted load and line operators over at most 19 reports. Load ON severity is profiled within the original bounds; an accepted OFF changes the operator while retaining the inferred state.

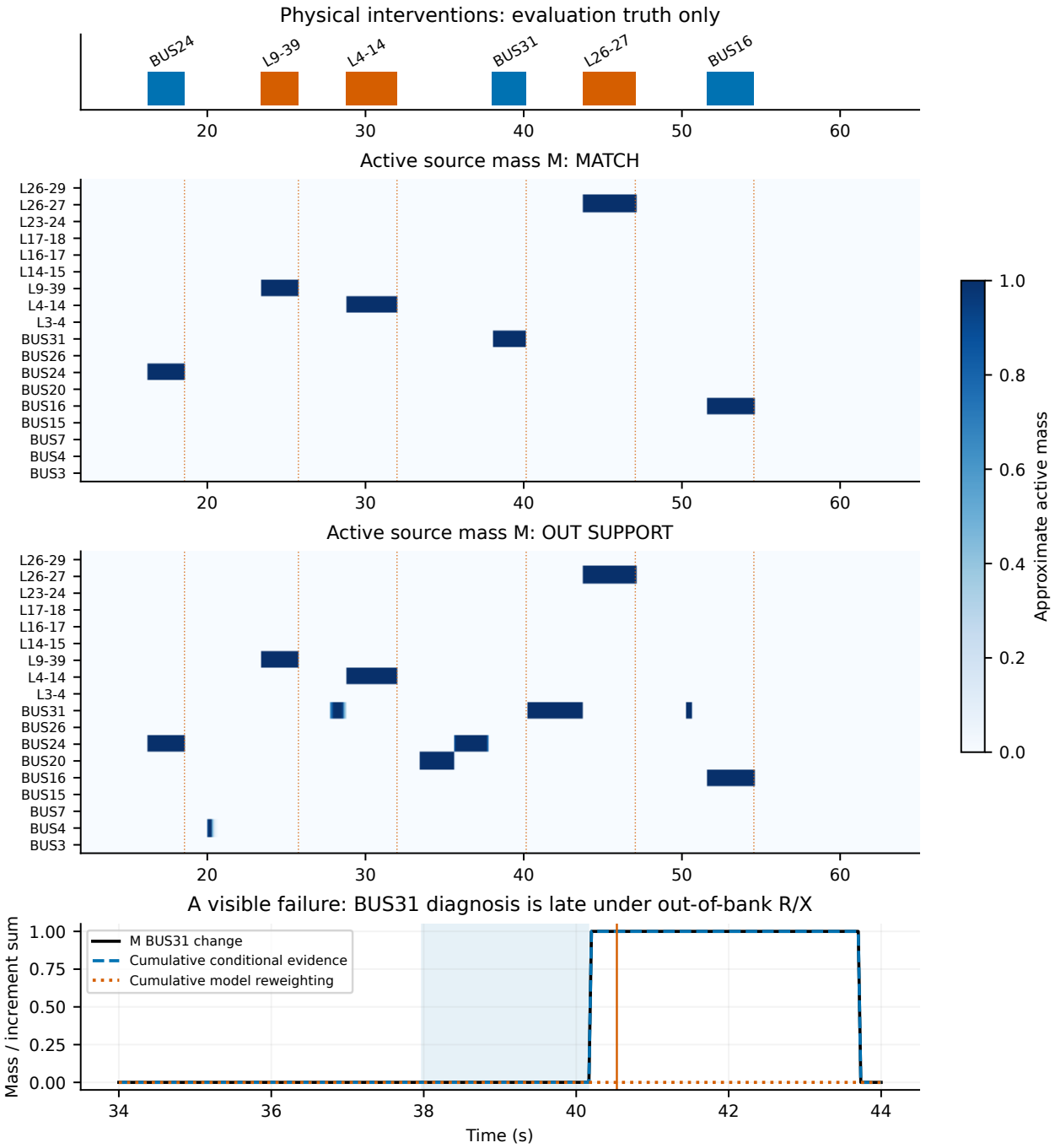


FIGURE 6. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

TABLE 5. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

Alternative histories then expire. This is a complete simpler procedure, not a deliberately memoryless reset baseline.

The single-state procedure and nominal history bank share fitted nominal operators; the five-model bank additionally mixes the declared resistance and reactance alternatives. All use a six-report source-stability clock. The two archived history-bank outputs are republished with that clock. The GLR includes a fixed penalty for its maximum search count; neither this penalty nor a numerically shared threshold makes its scores posterior probabilities or equalizes false-alarm rates. Table 5 aggregates all twelve records. The five-model bank obtains 51/54 timely decisions with 3 extras, versus 37/54 and 37 for the single-state procedure. This favors the full procedure in this development contrast, but does not isolate history averaging from severity integration, search or regime selection. These are three event calendars with repeated conditions. The zero alarms over 204 normal seconds are too little exposure to establish a low operational alarm rate. The changed counts relative to the original publication policy also show that the decision clock is consequential. The corresponding orphan-restoration counts are 32, 5 and 3 for the single-state, nominal and five-model procedures; zero premature matched closures does not imply error-free restoration. The original publisher gave 53/54 with 4 extras for the mixture and 54/54 with 7 for nominal. The common publisher gives 51/54 with 3 and 44/54 with 6, respectively. These are different decision policies on the same H04 outputs, not conflicting estimates of one endpoint.

B. WHAT GROUPED SIM TRAINING ESTABLISHES

We also fit two ExtraTrees variants using dependency groups, rather than randomly splitting correlated reports. The approximately 70/30 split is subdivided into 31 fitting, 8 validation and 17 reserved groups. SHORT is selected on validation and frozen before the reserved evaluation. Table 6 retains its marked load/line asymmetry. Its 302/576 timely reserved decisions and 3 extras do not establish adequate general transfer. The corpus was already open and all physical sources remain represented in development; this is neither a blind test nor an unseen-source split. Only one reserved normal dependency group underlies the 3696 normal seconds with 0 alarms.

Replaying the selected classifier on the common H04 cohort gives 27/54 timely diagnoses and 3 extras: 0/27 loads and 27/27 lines. The physical bank's benefit therefore lies chiefly in loads in this contrast; the classifier resolves all tested lines. Its frozen publisher uses six identical argmax labels, without the physical procedures' score threshold. This is a common-cohort, common-metric comparison of complete procedures, with different publishers. H04's normal parent also appears in the reserved SIM fraction and is not additional independent normal evidence.[7]

TABLE 6. Grouped SIM development. SHORT is selected only on validation; the reserved 30% is evaluated once. These cohorts are distinct from H04.

Variant and partition	Load	Line	Extras
Baseline, validation	5/96	87/96	0
SHORT, validation	16/96	92/96	0
SHORT, reserved	28/288	274/288	3

C. MEASUREMENT INTEGRITY AND RAW0001

The frozen physical filter does not admit RAW0001: the initial prefix violates its ROCOF noise-scale and temporal-stationarity support. The following ZIP and segmented-ML replays therefore examine complementary integrity and source interfaces; they do not validate R11 transfer. The submitted multi-family artifact is replayed on all 180 fixed windows of RAW0001. Its predictions are emitted at window completion. A separate availability monitor reports missing PMUs from received flags and finite-value masks; it keeps missing-PMU identity, corrupt-PMU identity and physical source distinct. Such diagnoses do not establish malicious intent. Complete-window corruption diagnoses describe their observed interval and are not broadcast backwards as causal labels.

Against the current annotation support, the submitted artifact detects an abnormality in 21/39 positive windows, with 1 false-positive window among 141 normal windows; all 180 windows produce outputs. Correct physical-family coverage is 4/24. The availability monitor identifies all 9200 missing reports and their PMU, whereas corruption localization is correct in 1/2 positive windows. A separate spurious corruption window is retained. These denominators describe one retrospective record, not independent field trials. Both implementations identify all 4 missing intervals and their PMU; the immediate interface reduces mean first-notice delay from 12.13 to 0 seconds on the received-report clock. This limited publication benefit does not measure wall-clock latency or physical inference. Its event-code annotations do not certify individual physical actuation boundaries, so actuation delay and restoration are not scored.

The archived source spreadsheet permits a separate metadata-based source comparison. A frozen hybrid uses the SIM-trained SHORT classifier only to replace the submitted artifact's load/line source when a first same-family diagnosis has already been emitted within that window. Predicted mixed physical/missing cases are eligible under the same rule. All classes, integrity diagnoses and window emission times remain fixed. Missing input splits the ML stream into complete segments with explicit unavailable startup; their

[7]Drafting/revision assistance: OpenAI Codex [1].

TABLE 7. Retrospective RAW family-and-source correctness on every physical-positive window, using the archived source metadata and current event-code support. Multiple windows from a sustained annotation are not independent actions.

Family	Submitted ZIP	Source hybrid
Fault	1/1	1/1
Line	0/1	1/1
Generation	0/11	0/11
Load	0/11	0/11
Total	1/24	2/24

prefixes are not verified normal. This is a segmented ML transfer test, not a successful execution of the physical history bank. Joint correctness changes from 1/24 to 2/24: 1 paired gain and 0 losses, with 179/180 total windows retaining the ZIP source. The gain corrects the line location; load and generation remain unresolved. A single improved window in one previously used record does not establish reliable all-family transfer.

The implemented interface combines these physical and integrity outputs in one chronological stream. Exact projection checks recover all 180 component windows and the availability state across 161379 reports. This verifies composition without adding a performance gain; completed-window evidence cannot reopen an already restored missing-PMU episode.

The calendar recovered for the archived hackathon table does not match the actual RAW annotations; its score is not a reproduced comparator. Relabeling windows or relaxing one admission gate would not validate the physical/instrument model. The archive preserves these failures and the current-calendar evaluation. Superiority over the historical all-family system remains unestablished.

VIII. REFERENCE AND RESPONSE ARE DIFFERENT CORRECTIONS

A separate controlled experiment tests a component of the local predictor: whether correcting the PMU reference is sufficient when operation changes. It compares a fixed signature, that signature with the measured prefix offset, and an operating-conditioned signature with the same offset. All methods receive the load family and ON/OFF times. Conditioning reduces median relative severity error, evaluated at the generating source, from 1.2% to 0.329%, while matching reference-only source accuracy. Thus better response fidelity does not automatically improve source decisions, and these oracle-context results are not another unknown-onset detector score. The supplementary material gives the mixed derivative, full scenario design, paired results, nominal-condition counterexample and signal plots. This component experiment motivates conditional signatures without enlarging the causal stream claim.[8]

IX. INTERPRETATION AND LIMITS

[8]Drafting/revision assistance: OpenAI Codex [1].

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{15}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{16}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [10], [11].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. RECOVERY IS NOT A UNIVERSAL SOURCE-INFORMATION GAIN

A separate full-F01 control supplies the prior history, later onset, family and candidate source pair. Profiled source sep-

aration is essentially unchanged under the known history and decreases when the tested prior-severity uncertainty is added; the binary source comparison gains no mean accuracy. The full scenarios, figure and numerical results are retained in the supplementary recovery control. This negative result supports retaining recovery for prediction consistency, without claiming that it universally adds source information or proves an R11 performance gain.

C. OPERATIONAL MEANING AND SCOPE

The physical distinction underlying the stream method is simple: restoration changes an input or topology, but does not erase the machine state. Retaining that state lets recovery compete with a new intervention. The development comparison also shows why alarm count and timely location must be reported together. Stricter admission can suppress extras while rejecting a real load. The existing admission ablation does not isolate the contribution of every recovery-memory operation.

The signature experiment addresses a different source of error. Its corrected variants receive the same pre-event measurements and offset. Any difference between them therefore comes from the event–context term and its estimated context, rather than a privileged baseline or longer window. This isolates the tested response approximation, but not an improvement of the unchanged causal detector. Known timing, one scalar context and four load sources make it a controlled component test.

Our construction builds on trajectory-based diagnosis, sequential likelihoods and multiple-model filtering. Their general principles are not new contributions here. The methodological claim concerns the specified competition among intervention and recovery histories; the experiments measure its operational behavior. Numerical superiority is claimed only where a common-data comparison supports it, not over Warner or other procedures that were not executed.[9]

The continuous physical catalog contains nine load buses and nine lines and requires an intervention-free startup. Its operation through missing PMUs, unknown families, source-excluded causal transfer and field conditions is unvalidated. The separate retrospective integrity interface in Section VII does not extend that physical catalog. F01 retains a provisional BUS5 current terminal and excludes BUS39 current angle. Agreement with this research model does not establish the producer's CT placement or exact reproduction of RAW0001. Local predictors, history pruning, covariance approximations and fitted startup parameters also limit probability claims.

F01's noise port applies temporal and cross-PMU structure within each signal family. Its archived cross-family factor covariance is not applied, and frequency/ROCOF noise is added after phase estimation. The inference comparison therefore concerns this declared simulator, not the complete dependence structure of a physical PMU installation.

[9]Drafting/revision assistance: OpenAI Codex [1].

The two ExtraTrees adapters are particular frozen implementations. Their load-transfer failure does not rule out learned load diagnosis or establish a better official hackathon score. Chen's eligible test concerns persistent line openings; its F01 adapter has DC/white-injection assumptions that do not capture all instrumental effects. Its mismatches and censored decisions remain part of the comparison.

Joint context uncertainty, unrestricted simultaneous sources and calibrated posterior probabilities remain unvalidated. Such extensions must preserve source-discriminating response directions and account for dependence between estimation error and measurement noise.

Finally, processing reports causally is not proof of real-time throughput. Observation time, computation and queue delay must be measured separately. The present implementations remain research inference services rather than validated protection or control systems.

X. CONCLUSION

We formulated sparse-PMU event attribution through competing physical histories. Load/line predictors start from estimated past states and retain the cause of the response after restoration. Active mass M and its increment ΔM expose the changing explanation rather than replacing the physical dynamics.[10]

Under the common development publication clock, the mixture obtains 51/54 timely family/source decisions with 3 extras, versus 37/54 and 37 for the simpler state-retaining physical control. Independent confirmation remains in progress.

The frozen physical procedure rejects the RAW startup contract. The separate integrity and learned-source interfaces provide only limited retrospective improvements, preserving the transfer boundary.

The result is a specified physical attribution procedure and a testable separation of recovery, reference error and source discrimination. Its scope remains the declared simulation contracts. It does not establish all-family diagnosis, universal source recovery, calibrated probabilities or real-time deployment.

ACKNOWLEDGMENT

OpenAI Codex [1] assisted with drafting and restructuring the abstract, all numbered main sections, author biographies and supplementary derivations, protocols and examples, LaTeX preparation, and plotting and execution-infrastructure code. The authors retain responsibility for the scientific content, verification, and final submission.

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JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.[11]

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.

MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.

PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.

DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.

[11] Biographical drafting assistance: OpenAI Codex [1].

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Supplementary Material for Physics-Based Event Attribution From Sparse PMUs With Recovery Memory

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT This supplement gives the executable statistical contract of the causal physical-history method: local DAE reduction, candidate transitions, truncated-severity evidence, state updates, restoration, worked examples, and scenario-level reproduction. It also defines the measurement-integrity interface and separates causal alarm, diagnosis and restoration metrics from the historical hackathon window evaluation. Earlier reconstruction, batch-source and proposed joint-estimator studies are preserved in the separately compiled Historical Technical Notes. Their results do not enter the causal method's denominators.

INDEX TERMS phasor measurement units, physical event attribution, measurement integrity, causal inference, reproducibility.

APPENDIX A HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections A–C give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.[1]

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal history bank, archived as R11. The conditional Gaussian updates use standard filtering identities [2]; the derivation below specifies their role and the approximations made by the implemented bank.

[1]Drafting/revision assistance: OpenAI Codex [1].

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the frozen method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the frozen operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current

discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.

3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The frozen screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.
7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the frozen detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{1}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FROZEN NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1-\exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

Active mass 0.99 for six reports confirms detection; restoration mass 0.99 for 15 reports confirms OFF. Source/line confirmation and episode replacement additionally follow the archived publication configuration. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their frozen publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

APPENDIX B LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e-sa)^\top S^{-1}(e-sa) + (a-\mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a-\mu')^2/v', \end{aligned} \tag{2}$$

$$v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \tag{3}$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \end{aligned} \tag{4}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l-\mu')/\sqrt{v'}$, $\beta = (u-\mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \tag{5}$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (4) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. The script scripts/verify_

supplement_examples.py checks the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a frozen experiment.[2]

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$S_o = R + J_o P_\rho J_o^\top, \qquad K_o = T_o P_\rho J_o^\top S_o^{-1},$$
$$m_h^+ = T_o m_h + g_o + K_o e, \quad d_h^+ = T_o d_h + u_o - K_o s. \tag{6}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. R11 does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa})e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \tag{7}$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\Delta M_c = \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \tag{8}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX C SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The active simulator is the frozen research candidate `SIM0001_JuliaPD_F01`, not an officially validated equivalent of RAW0001. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the frozen impedance law.[3]

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. RAW0001 provided retrospective calibration context. RAW0002 remains outside these studies.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

H03 and H04 each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

C01 prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18 + 27 + 12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. At this document snapshot, complete confirmation artifacts have not been imported; these counts specify the intended design and are not completed performance results.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON

[2]Drafting/revision assistance: OpenAI Codex [1].

[3]Drafting/revision assistance: OpenAI Codex [1].

alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. Chen2016 is eligible on that lane; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The E_i/X02 signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The X04 recovery geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate R11, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's frozen plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the F01 state, parameter and measurement files from its snapshot. For any new simulation create a new run directory; a new hypothesis requires a successor candidate. Do not run historical preparation scripts inside a frozen baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures with scripts/build_manuscript_bundle.ps1. Its default compilation does not regenerate scientific artifacts or import new campaign results. An explicit artifact refresh is a separate operation.

The repository keeps the current walkthrough in this appendix, the frozen campaign evidence under `generated/frozen/`, and the earlier mathematical development in the separately compiled Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX D MEASUREMENT INTEGRITY AND THE HISTORICAL SUBMISSION INTERFACE

An electrical intervention and a measurement failure are separate objects. The physical output identifies a family and asset; the integrity output identifies missing or corrupt PMU observations. Missingness alone does not establish malicious intent. A concurrent physical event and missing PMU can therefore have two different locations.[4]

The successor interface reuses the fitted detector, classifier and localizer from the actual submitted archive, rather than a similarly named exploratory pipeline. Archive member hashes are checked before inference. The input schema admits timestamps, three-phase voltage/current magnitudes and angles, frequency, ROCOF and the observed availability flag. Event labels, locations and annotated ON/OFF times are excluded. The fitted feature lists are also checked for label fields. This distinction matters because a historical diagnostic utility estimated a baseline from labeled normal samples over the complete record; that utility is not part of this inference interface.

The submitted classifier uses a physical-family head, a bad-data head and a missing/composition head with fixed rules. Its E5 and E7 outputs refer to missing and bad-data PMUs, respectively. E6 denotes missing data together with a physical prediction; its legacy location is the physical source, not the missing PMU. In the archived configuration, both composition and physical-confidence thresholds are zero, so E6 requires a nonzero physical prediction in a window with missing data; it is not a calibrated test of concurrent causation. E8 is the legacy mixed-physical proxy class, not an open-set rejection guarantee.

The availability interface evaluates each received report without looking ahead. It retains the affected PMU and channel mask rather than replacing missing measurements with future samples. The historical bad-data classifier uses a complete window and reports its result at the end. Within-window interpolation and centered features are permitted only under that delayed decision contract. Broadcasting the result to earlier timestamps does not make those earlier labels causal. The inherited cyber/integrity component is a comparator and integration component, not a newly claimed detector.

The implemented composition has two temporal scopes. Received-report availability messages update the current missing-PMU state. Completed-window records describe only their observed intervals; they retain separate physical and integrity fields and cannot assert current corruption or physical restoration. Their joint emission is the later compo-

[4]Drafting/revision assistance: OpenAI Codex [1].

nent emission, with each original time retained. For example, a window that contains missing data can finish after the PMU has resumed reporting; publishing that interval must not reopen the missing episode. Likewise, an earlier ML source diagnosis does not advance the submitted classifier's window-completion time.

The sealed stream contains 206 messages, including 180 completed windows. Post-seal projection checks reproduce the physical source, integrity diagnosis, source-selection provenance and unavailable states of the components, and recover current availability on all 161379 reports without disagreement. The comparison retains the original component evaluations. It neither reruns inference nor creates a combined accuracy from report-level missingness and window-level physical diagnosis. The frozen physical filter remains explicitly unavailable.

APPENDIX E ALARM, DIAGNOSIS AND RESTORATION ARE DIFFERENT DECISIONS

Let τ and ω be true ON and OFF, and let t_A, t_D and t_R denote the emission times of an alarm, source diagnosis and restoration. The detector can signal before it has a source. An eventual correct source at $t_D > t_A$ is not a correct location already available at t_A. Timely attribution requires a correct family and source at $t_D < \omega$. Estimated onset times never replace emission times in a delay metric.[5]

An event is associated with the first eligible unmatched alarm using time, before inspecting the predicted label. Incorrect labels remain associated errors rather than being reassigned to another event to improve the score. An alarm without a diagnosis, a wrong diagnosis, and no alarm are distinct outcomes. Restricted delay includes failures up to the evaluation horizon; conditional delay among successes is reported separately.

A restoration of an unmatched episode and a premature restoration of a matched event are also different errors. With a tolerance δ fixed before scoring, premature restoration is $t_R < \omega - \delta$; its lost active duration is $[\omega - t_R]_+$. Signed restoration delay is $t_R - \omega$. Report its distribution, missing closures and episodes still open at the horizon; a median alone can hide early closures.

The original C01 evaluator is preserved. Its false-restoration counter counts restorations of unmatched episodes, and its first-correct field can use the diagnosis associated with the first alarm even when emitted later. The successor evaluation reports these quantities with their restricted meanings and adds explicit premature closure and joint-decision metrics. This is a correction of evaluation semantics, not a change to the frozen detector or evidence that either failure occurred in a particular C01 record.

[5]Drafting/revision assistance: OpenAI Codex [1].

APPENDIX F RETROSPECTIVE RAW COMPARISON AND INDEPENDENT UNITS

The archived RAW0001 table contains 21 contexts, including nine labeled-normal and twelve abnormal contexts. Its recovered loader places a 30-s window three seconds before a supplied abnormal onset, and selects normal contexts from supplied normal intervals. Even a successful reproduction would measure classification and localization within event-guided contexts, not continuous detection. However, the recovered calendar is incompatible with the event annotations in the actual RAW files. File-history checks show that those RAW files have not changed, and the timing differences cannot be corrected by a common offset. The archived score is therefore not a reproduced comparison endpoint. The unsuccessful reconstruction and its provenance audit are retained; predictions from mismatched windows are not counted as accuracy against the archived labels.[6]

A separate complete-record replay uses fixed windows without event-guided placement and retains all normal, physical, missing and corrupt periods. Both interfaces preserve availability failures and abstentions. Physical and integrity locations are scored separately where their meanings differ; the legacy single-label export is retained for comparison. Physical-source scoring additionally requires verified source annotations; an event-code sequence alone cannot identify every intervention boundary or source. The separately sealed successor analysis uses the recovered original source workbook for physical identity, retaining current event-code support for window membership. Its coarse timing does not resolve the individual actuation or restoration times. This replay is not an official competition score. RAW0001 previously informed F01 calibration, so even a frozen new replay remains retrospective.

Simulation partitions use the complete physical parent and all of its derived windows, conditions and noise realizations. Validation belongs inside the development fraction. Repartitioning previously inspected campaigns does not restore independent-test status. Uncertainty calculations resample physical parents, retaining their paired conditions together. Different PMU channels and adjacent 30-Hz reports are not independent trials.

APPENDIX G REPOSITORY BOUNDARIES AND REPRODUCTION

The article and current supplement compile independently from `main.tex` and `supplement.tex`. The separate `historical_notes.tex` preserves corrected geometry and evidence theorems, earlier state reconstruction, the historical batch estimator, the proposed joint estimator and their original empirical qualifications. Moving this material does not remove negative results or upgrade an unexecuted estimator to an implemented one.[7]

[6]Drafting/revision assistance: OpenAI Codex [1].
[7]Drafting/revision assistance: OpenAI Codex [1].

The submission successor is versioned in the research repository under experiments/PAPER_SUBMISSION_S01/. It contains the source provenance, observation-only interface, metric specification, synthetic regressions, and separate historical-window and complete-record outputs. Performance tables enter the manuscript only after prediction sealing and evaluation. A successful code check or PDF build is not itself a performance result.

APPENDIX H EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.[8]

For a load ON alternative, write the innovation as $e_k - a s_k$. The affine conditional state is $m_k + a d_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{9}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{10}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{11}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a} U - \tfrac{1}{2} \widehat{a}^2 G - C_{\mathrm{ref}}. \tag{12}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c) P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The H04 comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX I GROUPED LEARNING AND INTEGRITY PUBLICATION

A. RETROSPECTIVE SIM TRAINING CONTRACT

The SIM11–19 registry contains 176 observation records from 88 physical cases. Connected components join shared physical histories and observation replicas before splitting. Of 56 dependency groups, 31 are used for fitting, 8 for model selection and 17 for the reserved development evaluation: 90, 16 and 70 records, respectively. This implements the requested approximately 70/30 development/reserved partition with an internal validation subset. All campaigns were already open; the split is retrospective and does not establish unseen-source or external generalization.

Both ExtraTrees variants use 400 trees, minimum leaf size two, square-root feature sampling, balanced class weights and seed 2601026. Fitting uses every third report starting at index 360. The target is normal outside an intervention and one of 18 load/line sources during it; recovery is not a separate training class. The baseline has 846 trailing features; SHORT adds 611 features. Their input is the same 47-channel representation with a 300-report empirical reference, colored filtering and median/MAD scaling. No DAE fit or RAW annotation enters fitting. The empirical reference has no startup stationarity guarantee; it is distinct from the physical filter's admission gate.[9]

The publication adapter requires six consecutive winning source labels. Scores are not claimed to be calibrated probabilities. Selection maximizes validation diagnoses with correct family and source emitted before OFF, then minimizes extra alarms, preferring the baseline on a complete tie. SHORT is frozen after 108/192 versus 92/192 validation successes. There is no refit on validation and no subsequent

[8] Drafting/revision assistance: OpenAI Codex [1].

[9] Drafting/revision assistance: OpenAI Codex [1].

parameter or threshold selection from the reserved 70 records. The declared metric associates the first temporal alarm independently of its predicted class, within 30 seconds or before the next onset, whichever comes first; success also requires diagnosis before the current OFF.

The reserved result is 302/576, comprising 28/288 load and 274/288 line interventions, with 3 extra alarms. Only two distinct normal dependency groups exist: one in training and one reserved; validation has none. The 3696 normal seconds are repeated observations of the reserved normal group, with 0 alarms. They do not provide independent long-duration operational exposure. In particular, neither report-level binomial intervals nor a validation false-alarm rate would be justified. Load weakness and the common F01 ancestry limit transfer claims even when the group assignment is disjoint.

The selected model is also replayed on all twelve H04 records, using the same successor event metrics as the physical controls. It obtains 27/54 timely decisions, 3 extras, 3 orphan restorations and 27 missing closures, including missed events. All missed events are loads; all tested lines are diagnosed before OFF. Its six-argmax publisher has no confidence threshold, so this is not an identical decision-clock ablation. The three H04 event groups are disjoint from the SIM fitting, validation and reserved groups, but its single normal group is shared with the reserved group. The inference-only replay reads no truth, registry or split. An earlier roster contained unused event/normal metadata; it is preserved and documented. Removing that metadata reproduces every output array and event exactly on all twelve records, before separate post-seal scoring.

B. MISSING-PMU NOTICE TIMING

The continuous RAW replay contains 4 observed missing intervals. The exact submitted artifact and the immediate availability interface both identify all intervals and their affected PMU. First correct notices have mean delay 12.13 versus 0 seconds, respectively. The clock is the received report timestamp relative to the first affected report; zero means emission on that report, not zero computation or communication latency. Notices before onset cannot match a later missing interval, and a later correct source cannot repair an incorrect first notice. This is a publication benefit from already-observed flags and finite-value masks, not evidence of a new malicious-attack detector or improved physical-source inference.

C. SEGMENTED ML TRANSFER AND SOURCE COMPOSITION

The classifier consumes the phase-A voltage/current magnitudes and angles, frequency and ROCOF in the declared eight-PMU order. Complete segments require all 48 measurements finite and all presence flags true; missing reports are never removed to join their neighboring samples. Each segment gets its own 300-report empirical reference and remains unavailable through index 359. Segment startup is not certified normal and physical state is not preserved across gaps. The same frozen classifier and publication rule are used; no RAW source label, event code or score enters inference.

For a submitted-artifact prediction of line or load, including its predicted physical component in a mixed missing case, source replacement requires the first ML diagnosis of an episode with that same predicted family, emitted inside the completed window no later than the artifact's own emission. Later revisions cannot repair that first diagnosis. The earliest eligible diagnosis is selected; otherwise the ZIP source is retained. Fault, generation, integrity outputs and all window class labels and times are unchanged. The archived Top-3 list remains identified as the original ZIP list; it is not presented as the hybrid's reranked posterior.

The completed replay has 5 complete segments and 1 selected source substitution, with 179 fallbacks. The evaluator verifies full report coverage, explicit startup unavailability, causal message times and every replacement against the frozen rule before scoring. Source identity is supplied only to this later evaluator by the original metadata workbook: fault BUS39, line 23–24, generation BUS2 and load BUS7. Current event-code support determines which fixed windows contain each family; the workbook's approximate times do not supply exact action onsets or restorations. All physical-positive windows, including generation outside the learned catalog, enter the denominator. Paired joint correctness is 1/24 versus 2/24. These are dependent windows of one retrospective record; no population-level significance or official competition improvement is inferred.

APPENDIX J DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.[10]

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios

[10]Drafting/revision assistance: OpenAI Codex [1].

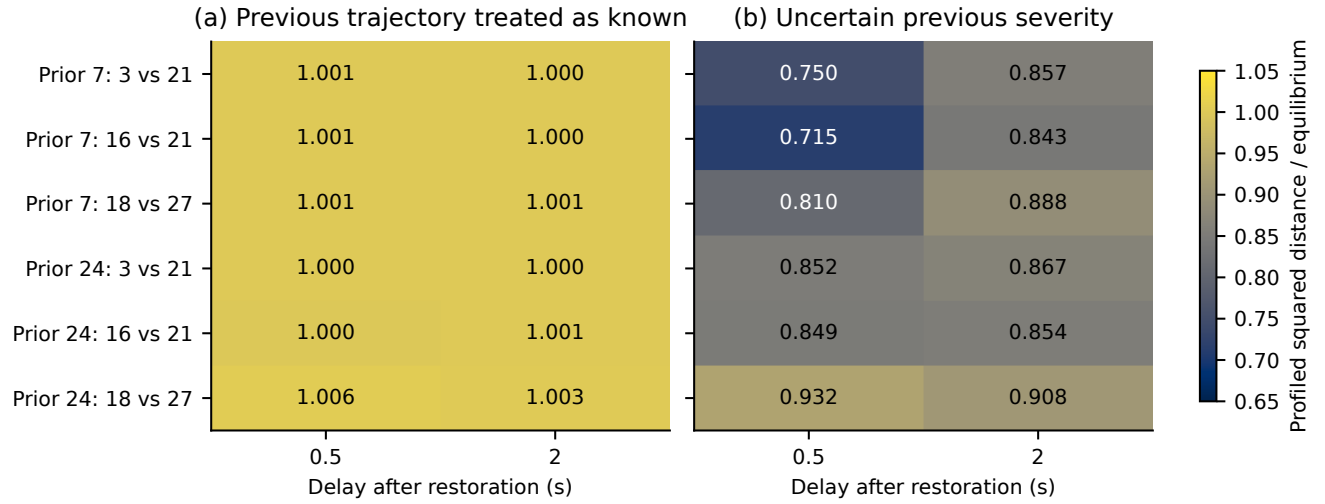


FIGURE 1. **Full-F01 physical control, not an R11 performance comparison. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

exceeds one (Fig. 1). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an R11 improvement nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

APPENDIX K OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of R11.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta,h) - y_i(\eta,-h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{13}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{14}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255

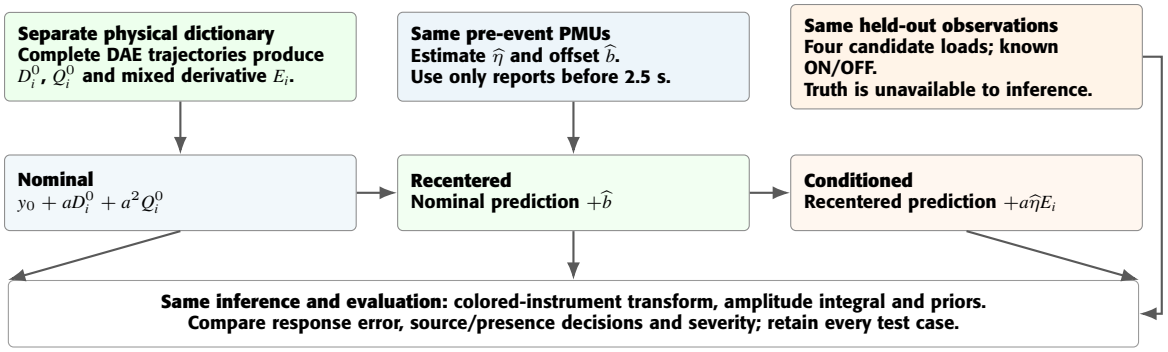


FIGURE 2. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

TABLE 1. Controlled known-onset load test. Response error uses 80 physical cases; joint and source-only correctness use 240 noisy event records. Severity error is evaluated at the generating source. The 15 normal windows are not normal-hours exposure.

Measure	Nominal	Recentered	Conditioned
Response error (%)	1.25	1.25	0.0561
Joint correct	114/240	240/240	240/240
Source-only correct	114/240	240/240	240/240
Normal selections	12/15	0/15	0/15
Severity error (%)	3.45	1.2	0.329

paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.[11]

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 3 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

ACKNOWLEDGMENT

OpenAI Codex [1] assisted with drafting and revision of the abstract and all numbered appendices, including the derivations, worked examples and protocols, and with LaTeX and code preparation. The section footnotes identify this writing assistance separately from scientific references. The authors retain responsibility for verification and the final manuscript.

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[2] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.

[11] Drafting/revision assistance: OpenAI Codex [1].

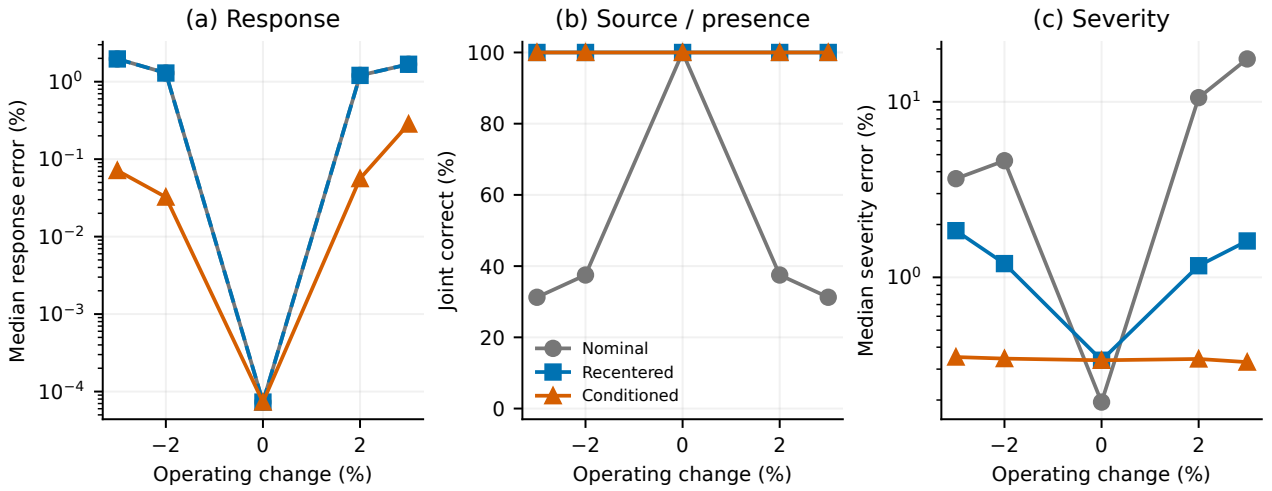


FIGURE 3. **Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.**

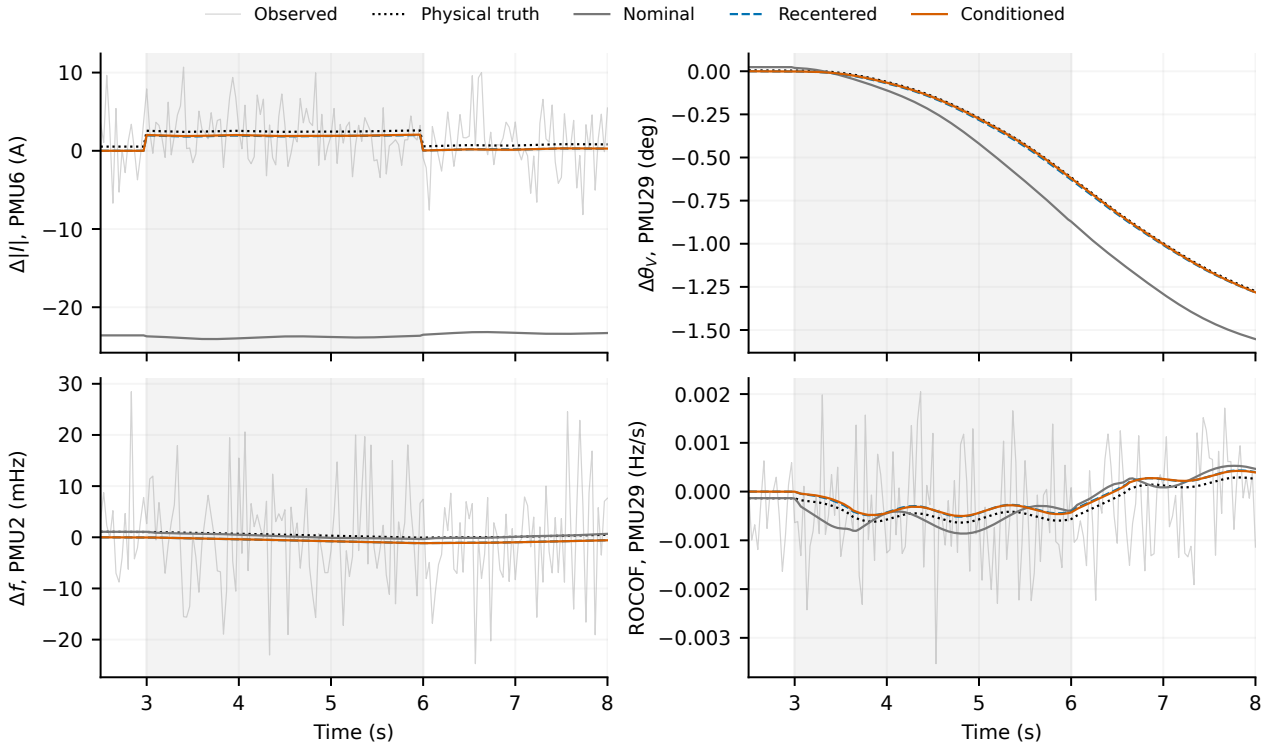


FIGURE 4. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**